Graph Colouring

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Graph colouring is an important concept in graph theory that deals with assigning labels (called colors) to elements of a graph subject to certain constraints. The most common type is vertex colouring, where colors are assigned to vertices in such a way that no two adjacent vertices share the same color.

Formally, given a graph $G = (V, E)$, a proper vertex colouring is a function $C : V \to \{1, 2, \cdots, k\}$ such that if u and v are adjacent vertices (i.e., $uv \in E$, then $c(u) \neq c(v)$.

Types of Graph Colouring:

Vertex Colouring – No two adjacent vertices have the same color.

Edge Colouring – No two adjacent edges (edges sharing a common vertex) have the same color.

Face Colouring – Used mainly in planar graphs (e.g., map colouring problems).

1 Vertex Colouring of Graphs

A simple graph is said to be **k-vertex colourable** if it is possible to assign one colour from a set of k number of colours to each vertex such that no two adjacent vertices are assigned the same colour. We say the graph is *coloured*.

The smallest number of colors required to properly color a graph G is called its **chromatic number**, denoted by $\chi(G)$.

Consider the following graph G:

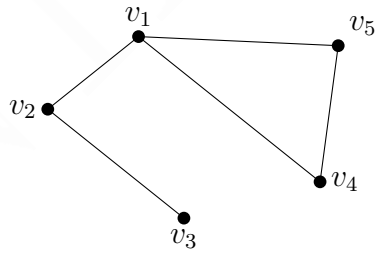


Fig. 1.1

Consider the set of four colours {Red, Green, Black, Blue}. If we assign Red to v_1, Green to v_2, Blue to v_3, Black to v_4 and Blue to v_5, then we see that no two adjacent vertices are assigned the same colour. So G is a 4-vertex colourable graph.

Again, we could assign Red to v_1, Green to v_2, Red to v_3, Black to v_4 and Green to v_5. So we see G is 3-vertex colourable also.

Note that it would be impossible to meet the above requirements by only two colours. Hence G is a graph with chromatic number 3 and $\chi(G) = 3$.

Note:

1. If a graph G has n-number of vertices then, $\chi(G) \leq n$.
2. If G is a single vertex graph then, $\chi(G) = 1$.
3. If G is a null graph then, $\chi(G) = 1$.

4. If a graph G has a loop at a vertex v then v is adjacent to itself and so, no colouring is possible for G. To avoid this situation, we remove all the loops if it has.

5. If there exists parallel edges joining two vertices of a graph, then remove all except one, because solution for the cases would be same.

6. If H is a subgraph of G then $\chi(H) \leq \chi(G)$.

Theorem 1.1. *The chromatic number of a complete graph with n vertices is n. i.e., $\chi(K_n) = n$.*

Proof. Let K_n be a complete graph with n vertices $v_1, v_2, \cdots, v_n$.

Let a colour C_1 be assigned to v_1. Since v_2 is adjacent to v_1, a different colour must be assigned to v_2. Let this colour be C_2.

Now v_3 is adjacent to both v_1 and v_2, so it cannot be assigned either C_1 or C_2. Hence a new colour C_3 is required for v_3.

Proceeding in this way, each vertex v_k $(k = 1, 2, \cdots, n)$ is adjacent to all the previously coloured vertices $v_1, v_2, \cdots, v_{k-1}$. Therefore, a new colour C_k must be assigned to v_k.

Thus, n distinct colours $C_1, C_2, \cdots, C_n$ are required to colour the graph K_n.

If any one of the colours is removed, then at least two adjacent vertices would receive the same colour, which is not allowed. Hence, K_n cannot be coloured with fewer than n colours.

Therefore, $\chi(K_n) = n$.

□

Theorem 1.2. *The chromatic number of a cycle with n vertices (C_n) is*

$$\chi(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

Proof. Let the cycle C_n have vertices $v_1, v_2, \cdots, v_n$ appearing in cyclic order.

Assign colour 1 to v_1. Since v_2 is adjacent to v_1, it must be assigned a different colour, say colour 2. Now v_3 is adjacent to v_2 but not to v_1, so we may assign colour 1 to v_3.

Proceeding in this way, we alternately assign colours $1, 2, 1, 2, \cdots$ to the vertices $v_1, v_2, v_3, \cdots, v_n$ respectively.

Case 1: n is even.

If n is even, the last vertex v_n will receive colour 2. Since v_n is adjacent to v_1 (which has colour 1), there is no conflict. Thus C_n can be coloured with only two colours.

Also, at least two colours are required because adjacent vertices cannot have the same colour. Hence,

$$\chi(C_n) = 2 \quad \text{if } n \text{ is even.}$$

Case 2: n is odd.

If n is odd, the last vertex v_n will receive colour 1. But v_n is adjacent to both v_{n-1} (which has colour 2) and v_1 (which also has colour 1). Hence a conflict occurs.

Therefore, a new colour, say colour 3, must be assigned to v_n. Thus at least three colours are required.

Hence,

$$\chi(C_n) = 3 \quad \text{if } n \text{ is odd.}$$

□

Theorem 1.3. *The chromatic number of a non-null graph is 2 if and only if the graph is bipartite.*

Proof. Let G be a bipartite graph. Then its vertex set V can be partitioned into two disjoint sets

$$V_1 = \{v_1, v_2, \cdots, v_m\} \quad \text{and} \quad V_2 = \{u_1, u_2, \cdots, u_n\}$$

such that every edge of G joins some vertex in V_1 to some vertex in V_2. Obviously, no two vertices within the same set are adjacent.

Assign colour 1 to each vertex in V_1 and colour 2 to each vertex in V_2. Since no two vertices in the same set are adjacent, and every edge joins a vertex of V_1 to a vertex of V_2, this colouring is proper.

Thus the graph can be coloured using only two colours. Hence, $\chi(G) = 2$.

Conversely, suppose $\chi(G) = 2$. Then the vertices of G can be coloured using exactly two colours such that no two adjacent vertices have the same colour.

Let V_1 be the set of vertices assigned with colour 1 and V_2 be the set of vertices assigned with colour 2. Then V_1 and V_2 form a partition of V, and no two vertices within the same set are adjacent.

Hence, G is bipartite. □

Theorem 1.4. *A tree with two or more vertices has chromatic number 2.*

Proof. Let

$$V = \{v_1, v_2, \cdots, v_n\}, \quad n \geq 2$$

be the vertex set of a tree T. Two subsets V_1 and V_2 are formed from V in the following way:

The vertex $v_1 \in V_1$. If v_2 is adjacent to v_1 then $v_2 \in V_2$. Otherwise $v_2 \in V_1$. If v_3 is adjacent to at least one vertex in V_1 then $v_3 \in V_2$, otherwise $v_3 \in V_1$. In this way each vertex of V is classified into the two non-empty classes V_1 and V_2 where

$$V_1 \cap V_2 = \varnothing$$

and

$$V_1 \cup V_2 = V.$$

Since no two vertices in V_1 are adjacent we can assign the same colour, say colour-1, to each member of V_1. Since no two vertices in V_2 are adjacent and every vertex in V_2 is adjacent to some vertex in V_1, we can assign another colour, say colour-2, to each vertex in V_2.

Thus T is coloured using only the two colours colour-1 and colour-2. Hence $\chi(T) = 2$. □

2 Upper and Lower bounds of chromatic number

2.1 Upper Bound

Let us define,

$$\Delta(G) := \text{maximum of the degrees of all the vertices of } G.$$
$$\delta(G) := \text{minimum of the degrees of all the vertices of } G.$$

Theorem 2.1. *For any graph G, $\chi(G) \leq \Delta(G) + 1$.*

Theorem 2.2 (Brooks Theorem)**.** *Let G be connected graph with $\Delta(G) \geq 3$. If G is not complete, then $\chi(G) \leq \Delta(G)$.*

2.2 Lower Bound

Definition 2.3. *Let $G = (V, E)$ be a graph and A be a subset of a vertex set. If no two vertices in A are adjacent in G then A is called* ***independent set of vertices****.*

In the Fig. 1.1, the set of vertices $\{v_1, v_3\}$ is independent set of vertices. The set $\{v_3, v_5\}$ is another independent set of vertices.

Definition 2.4. *The cardinal number of the largest independent set in a graph G is called the* ***independence number*** *of the graph, and is denoted by $\alpha(G)$.*

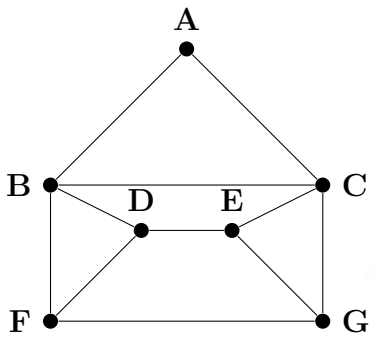


Fig. 2.1

In Fig. 2.1, the set of vertices $\{A, E, F\}$, $\{A, D, G\}$, $\{A, E\}$, $\{A, F\}$ are independent set of vertices and for this graph $\alpha(G) = 3$.

Theorem 2.5. *For a graph G with n-vertices, $\chi(G) \geq \dfrac{n}{\alpha(G)}$.*

Definition 2.6. *Let $G = (V, E)$ be a graph and A is a subset of V. If every pair of vertices in A are adjacent then A is called* ***clique*** *of G.*

Definition 2.7. *The cardinal number of the largest clique in a graph G is called* ***clique number*** *of G and is denoted as $\omega(G)$.*

Theorem 2.8. *For a graph G with n-vertices, $\chi(G) \geq \omega(G)$.*

Definition 2.9. *A* ***perfect graph*** *is a graph G such that for every induced subgraph H of G, the clique number equals the chromatic number.[i.e., $\chi(H) = \omega(H)$]*

3 Chromatic Polynomial

Definition 3.1. *The chromatic polynomial $P(G, x)$ of a graph G is the number of different possible colourings of the graph G with x-number of colors.*

Example 3.2. *Consider the complete graph G with three vertices A, B, C (i.e., a triangle) and the x number of colours $C_1, C_2, \cdots, C_x$. Since A, B, C are pairwise adjacent to each other, each of them should be assigned with distinct colour.*

Now A can be assigned with x number of colours; B can be assigned with $(x-1)$ number of colours and C can be assigned with $(x-2)$ number of colours. Therefore G may be coloured in $x(x-1)(x-2)$ ways with x colours.

Then, $P(G, x) = x(x-1)(x-2) = x^3 - 3x^2 + 2x$ is known as the chromatic polynomial of G.

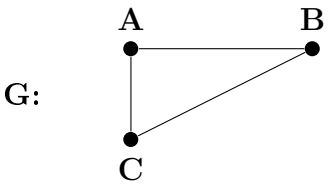


Fig. 3.1

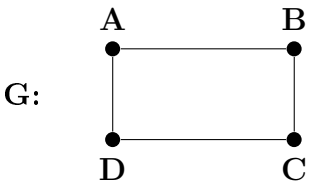


Fig. 3.2

Example 3.3. *Find the chromatic polynomial of the following graph:*

Suppose there are x-number of colors. The vertex A can be coloured in x-ways. Since B is adjacent to A, so it can be coloured in $(x-1)$-ways. Since, C is adjacent to B, so it can be coloured in $(x-1)$-ways.

Now, to color the vertex D, there are two cases arise:

Case 1: *Let C and A have the same color. After colouring A and B, there is exactly one way to color C, which is same as the color A. Now, D is adjacent to both A and C. So, D has $(x-1)$ choices to color. So, total colourings in this case is*

$$x \cdot (x-1) \cdot 1 \cdot (x-1) = x(x-1)^2.$$

Case 2: *Let C and A have the different colors. In this case, the vertex A has x-choices, vertex B has $(x-1)$-choices. Out of $(x-1)$ choices for C, one gives A and C have the same color. So remaining $(x-2)$ choices. Now the vertex D is adjacent to A and C, which now have different colors. So, D has $(x-2)$ choices. So, total colourings in this case is*

$$x \cdot (x-1) \cdot (x-2) \cdot (x-2) = x(x-1)(x-2)^2.$$

Finally adding both cases, we have,

$$\begin{aligned} P(G,x) &= x(x-1)^2 + x(x-1)(x-2)^2 \\ &= x^4 - 4x^3 + 6x^2 - 3x. \end{aligned}$$

Note: Chromatic polynomial of a cycle C_n with n-vertices is given by:

$$P(C_n, x) = (x-1)^n + (-1)^n(x-1).$$

Note. The smallest x for which $P(G,x) \neq 0$ is nothing but $\chi(G)$ because $\chi(G)$ is the least number of colours with which G can be coloured.

Theorem 3.4. *The chromatic polynomial of the complete graph K_n is*

$$P(K_n, x) = x(x-1)(x-2)\cdots(x-n+1) = \prod_{k=0}^{n-1}(x-k).$$

Proof. Let the vertices of K_n be $v_1, v_2, \cdots, v_n$. Suppose we have x available colours $c_1, c_2, \cdots, c_x$.

Since K_n is a complete graph, every pair of distinct vertices is adjacent. Therefore, no two vertices can receive the same colour in a proper colouring. Hence, all vertices must be assigned distinct colours.

We count the number of ways to assign colours:

- Vertex v_1 can be coloured in x ways.
- Vertex v_2 can be coloured in $(x-1)$ ways (any colour except the one used for v_1).
- Vertex v_3 can be coloured in $(x-2)$ ways.
- $\vdots$
- Vertex v_n can be coloured in $[x-(n-1)]$ ways.

Therefore, the total number of proper colourings is

$$P(K_n, x) = x(x-1)(x-2)\cdots(x-n+1).$$

□

Theorem 3.5. *The chromatic polynomial of a tree with n vertices is*

$$P(G, x) = x(x-1)^{n-1}.$$

Proof. Let G be a tree with n vertices. We prove the result by mathematical induction on n.

If $n = 1$, then G consists of a single vertex. It can be coloured in x ways. Thus,

$$P(G, x) = x = x(x-1)^0.$$

If $n = 2$, then G consists of two adjacent vertices v_1 and v_2. Vertex v_1 can be coloured in x ways. Since v_2 is adjacent to v_1, it can be coloured in $(x-1)$ ways. Hence,

$$P(G, x) = x(x-1).$$

Thus the formula holds for $n = 2$.

Assume that for every tree with m vertices, $P(G, x) = x(x-1)^{m-1}$.

Now let G be a tree with $m+1$ vertices. Since G is a tree, it has at least one pendant vertex. Let v be a pendant vertex and let v' be the unique vertex adjacent to v.

Remove v from G to obtain the graph $G - v$. Then $G - v$ is also a tree with m vertices.

By the induction hypothesis,

$$P(G - v, x) = x(x-1)^{m-1}.$$

Now consider extending a proper colouring of $G - v$ to G. Since v is adjacent only to v', it cannot receive the colour assigned to v'. Therefore, v can be coloured in $x - 1$ ways.

Hence,

$$P(G, x) = P(G - v, x)\,(x-1) = x(x-1)^{m-1}(x-1) = x(x-1)^m.$$

Thus the formula holds for $(m+1)$.

Hence by principle of mathematical induction, the chromatic polynomial of a tree with n vertices is $P(G, x) = x(x-1)^{n-1}$. □

Theorem 3.6 (Decomposition Theorem)**.** *Let u and v be two non-adjacent vertices of a simple graph G. Let G' be the graph obtained from G by adding the edge uv, and let G'' be the graph obtained from G by identifying (merging) the vertices u and v. Then*

$$P(G, x) = P(G', x) + P(G'', x).$$

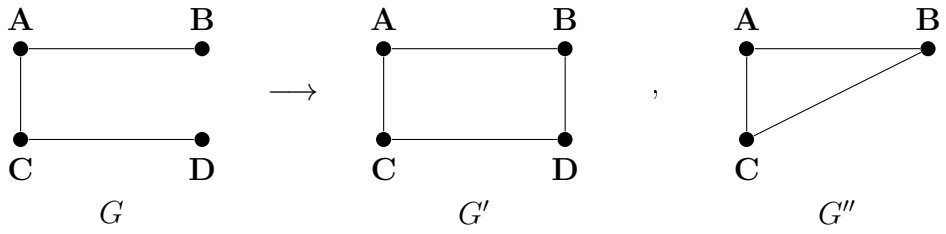


Fig. 3.3

Example 3.7. *Consider the following graph G and the graphs G' and G'' obtained from G. Using Decomposition theorem, find the chromatic polynomial of G'.*

Here,

- *G' is obtained from G by adding the edge between B and D.*
- *G'' is obtained from G by merging the vertices B and D.*

Since G is a tree with 4 vertices,

$$P(G,x) = x(x-1)^3.$$

Since G'' is isomorphic to K_3 so,

$$P(G'',x) = x(x-1)(x-2).$$

By the Decomposition Theorem, $P(G,x) = P(G',x) + P(G'',x)$, which implies,

$$\begin{aligned} P(G',x) &= P(G,x) - P(G'',x) \\ &= x(x-1)^3 - x(x-1)(x-2) \\ &= x^4 - 4x^3 + 6x^2 - 3x. \end{aligned}$$

Some properties of chromatic polynomials:

1. The degree of the chromatic polynomial $P(G,x)$ for a simple graph G with n-vertices is exactly n.
2. The leading coefficient is always 1.
3. For a graph G with n-vertices,
$$P(G,x) = x^n - a_1x^{n-1} + a_2x^{n-2} - \cdots + (-1)^n a_n,$$
where $a_i \geq 0$ and $a_n = 0$ always. That is constant term is always zero.
4. $P(G,0) = 0$.
5. $P(G,1) := \begin{cases} 1 & \text{if } G \text{ has no edge} \\ 0 & \text{if } G \text{ has at least one edge.} \end{cases}$

Example 3.8. *Find whether the polynomial $x^3 + 5x^2 - 3x + 5$ is a chromatic polynomial of some graph.*

Here, $P(G,x) = x^3 + 5x^2 - 3x + 5$. Then $P(G,0) = 5$. As, $P(G,0) \neq 0$, so it can't be a chromatic polynomial of any graph.

Example 3.9. *Find whether the polynomial $x^4 + 2x^3 - 4x^2$ is a chromatic polynomial of some non-null graph.*

A non-null graph G must have at least one edge. i.e., at least two vertices are adjacent. So at least two colors are needed for colouring the graph. That is $\chi(G) \geq 2$. So, $P(G,1) = 0$. But here, $P(G,1) = -1$. Hence, the given polynomial can't be a chromatic polynomial of any graph.

4 Edge Colouring of Graphs

Recall that two edges of a graph are called *adjacent* if they have a common end vertex.

A simple graph is said to be **k-edge colourable** if it is possible to assign a colour from a set of k colours to each edge in such a way that no two adjacent edges receive the same colour. In that case the graph is said to be *edge-coloured*.

The minimum number of colors required for edge-colourable of a graph G is called **edge chromatic number** of G and is denoted by $\chi'(G)$.

Consider the following graph:

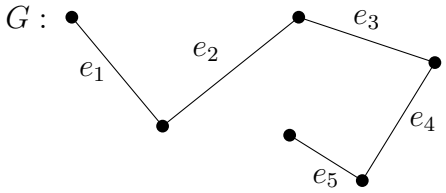


Fig. 4.1

For this graph, it is clear that, minimum two colors are required to edge-color the graph G. So, $\chi'(G) = 2$.

Theorem 4.1. *For a complete graph K_n with n-vertices ($n \geq 2$),*

$$\chi'(G) = \begin{cases} n-1 & , \text{ if } n \text{ is even} \\ n & , \text{ if } n \text{ is odd.} \end{cases}$$

Example 4.2. *Colour the edges of the following graph with the colour set $\{G, R, Y, B\}$. Find the edge-chromatic number.*

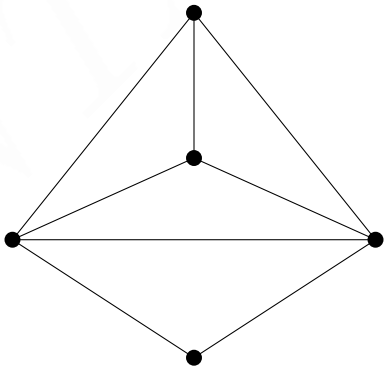


Fig. 4.2

Example 4.3. *Find the edge chromatic number of the following graph:*

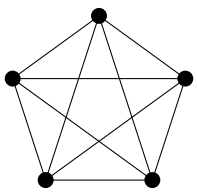


Fig. 4.3

This graph is a complete graph with 5-vertices. Since 5 is odd, so edge-chromatic number is 5.

Example 4.4. *A complete graph consists of 45 number of edges. Find the edge chromatic number of the graph.*

Let G be the complete graph having n number of vertices. Then, number of edges is $\frac{n(n-1)}{2}$.

By given condition, $\frac{n(n-1)}{2} = 45$ or, $n(n-1) = 90$ which gives $n = 10$.

So, G is a complete graph with 10 vertices. So, $\chi'(G) = 10$.

5 Face Colouring of Graphs

A map is called **face colourable** with a set of colours if we can assign one colour to each face (or region) of the map so that no two adjacent faces (i.e., faces sharing a common boundary edge) have the same colour.

A map is called k–**face colourable** if it can be coloured with k colours but cannot be coloured with $k-1$ colours.

Example 5.1. *Consider the following map.*

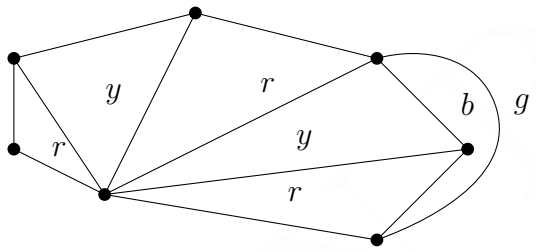


Fig. 5.1

*This map is 4–**face colourable** because its regions can be coloured with four colours $\{r, y, b, g\}$ such that no two adjacent regions have the same colour.*

Four and Five Colour Theorem:

The idea of the *Four Colour Theorem* was first proposed in 1852 by Francis Guthrie while he was trying to colour the map of the counties of England. He observed that only four different colours seemed sufficient to colour any map so that no two neighbouring regions share the same colour.

For many years mathematicians tried to prove this conjecture. Finally, in 1976, Kenneth Appel and Wolfgang Haken proved it using computer assistance. The result is now known as the *Four Colour Theorem*. In graph theoretic form it can be stated as follows.

Theorem 5.2 (**Four Colour Theorem**). *Every planar graph has chromatic number at most four.*

In fact, the *Five Colour Theorem*, stated below, also implies that planar graphs can be coloured with a small number of colours. The proof of the five colour theorem is considerably easier than that of the four colour theorem. Alfred Kempe attempted a proof of the four colour theorem in 1879, but later Percy John Heawood found an error and subsequently proved the Five Colour Theorem.

Theorem 5.3 (**Five Colour Theorem**). *Every planar graph can be coloured with at most five colours.*

Example 5.4. *Find the chromatic number of the following graph:*

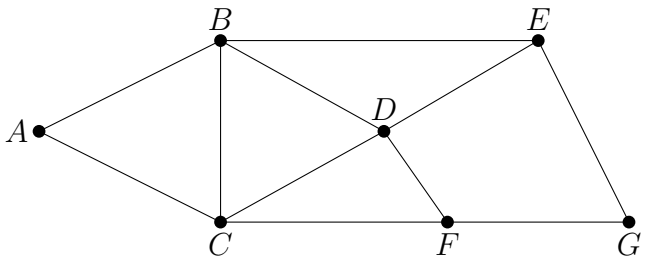


Fig. 5.2

Solution. Vertices A, B, and C form a triangle, so at least three colours are required. Assign colours as follows:

$$A \to \text{Red}, \qquad B \to \text{Blue}, \qquad C \to \text{Green}.$$

Vertex D is adjacent to B and C, so it cannot take Blue or Green. Hence D is coloured Red.

Vertex F is adjacent to D and C, so it cannot take Red or Green. Thus F is coloured Blue.

Vertex E is adjacent to B and F, so it is coloured Green.

Finally, vertex G is adjacent to E and F, so it can be coloured Red.

Thus the graph can be coloured with three colours, but not with two colours. Therefore, the **chromatic number** of the graph is $\chi(G) = 3$.

Example 5.5. *Find the chromatic number of the following graph:*

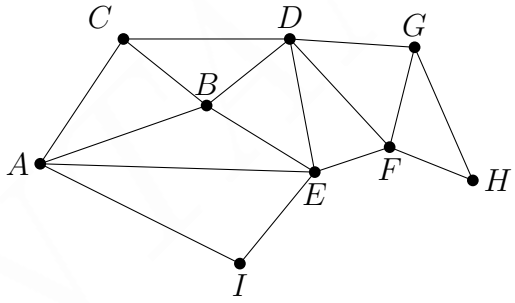


Fig. 5.3

Solution. Do it yourself. [Chromatic number=4.]

Example 5.6. *Label the vertices with colours for the following graph. Hence determine the chromatic number.*

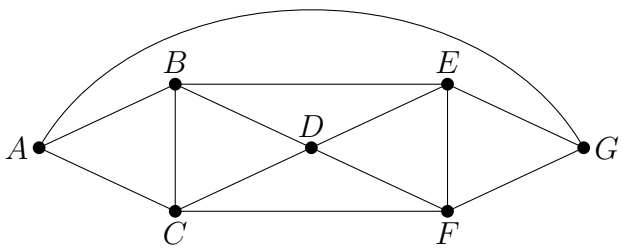


Fig 5.4

Example 5.7. *Find the chromatic number of the following graph (Fig. 5.5):*

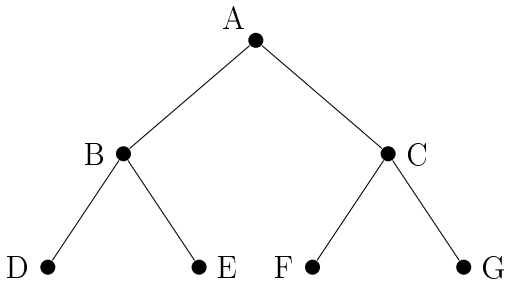


Fig. 5.5

Solution. The given graph is a tree. So, chromatic number is 2.

Example 5.8. *Consider the complete graph K_6 with six vertices a, b, c, d, e, and f. The graph G_1 is obtained from K_6 by deleting the edge ab. The graph G_2 is obtained from G_1 by further deleting the edge ed. Find the chromatic numbers of G_1 and G_2.*

Solution. Do it yourself. [Ans: $\chi(G_1) = 5$ and $\chi(G_2) = 4$.]

Example 5.9. *Find the chromatic number of the following graph. Find whether this graph is perfect.*

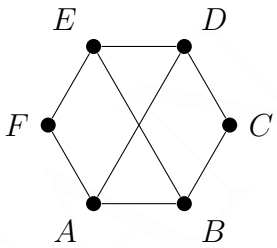


Fig. 5.6

Solution. Let the vertex A be assigned a colour say Green. Since B, D and F are adjacent to A so they should be assigned with another colour say Blue.

Since E and C are not adjacent to A they are assigned Green again. Therefore, chromatic number is 2.

A graph G is said to be perfect if for every induced subgraph H of G,

$$\chi(H) = \omega(H),$$

where $\omega(H)$ denotes the size of the largest clique of H.

Since the given graph can be coloured using only two colours, it is a bipartite graph. A bipartite graph contains no odd cycle and its largest clique consists of only two vertices. Hence

$$\omega(G) = 2.$$

Therefore,

$$\chi(G) = \omega(G) = 2.$$

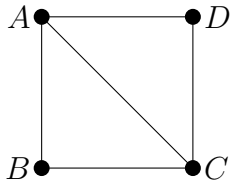
Also every induced subgraph of a bipartite graph is bipartite and satisfies the same property. Hence

$$\chi(H) = \omega(H)$$

for every induced subgraph H of G.

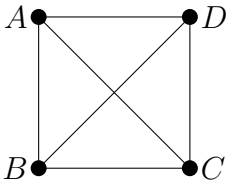
$\therefore$ The graph G is a perfect graph.

Example 5.10. *Using Decomposition theorem, find the chromatic polynomial of the following graph and hence find chromatic number.*

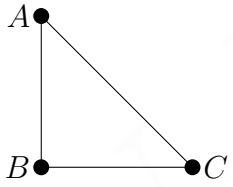


Solution. Let, G be the given graph. We obtain the graph G' be adding the edge BD and obtain the graph G' by merging the vertices B and D as follows:

G':



G'':



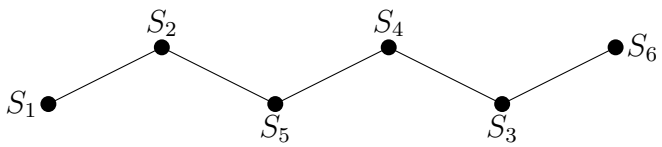
Since, G' is K_4 so, $P(G', x) = x(x-1)(x-2)(x-3)$.
Since, G'' is K_3 so, $P(G'', x) = x(x-1)(x-2)$.
By, Decomposition theorem,

$$\begin{aligned} P(G, x) &= P(G', x) + P(G'', x) \\ &= x(x-1)(x-2)(x-3) + x(x-1)(x-2) \\ &= x(x-1)(x-2)(x-3+1) \\ &= x(x-1)(x-2)^2. \end{aligned}$$

Now, $P(G, 1) = 0, P(G, 2) = 0$, but $P(G, 3) \neq 0$. So, $\chi(G) = 3$.

Example 5.11. *A new flag is to be designed with 6 vertical stripes using 4 colours. In how many ways can this be done so that no two adjacent stripes have the same colour?*

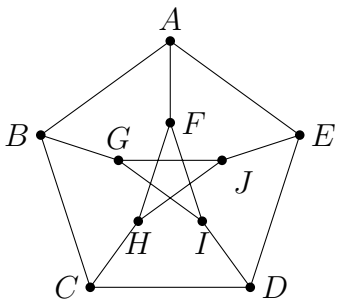
Solution: Let S_1, S_2, S_3, S_4, S_5, and S_6 be the stripes. The problem is nothing but the colouring of the following graph with four colours:



Clearly, this graph is a tree with 6 vertices.
The chromatic polynomial of this tree is $f(x) = x(x-1)^{6-1} = x(x-1)^5$.
Number of ways of colouring with at most 4 colours is $f(4) = 4(4-1)^5 = 972$.

Example 5.12. *Using Brook's Theorem find the chromatic number of the following graph (Petersen Graph)* Γ.

Γ:



The graph Γ *is neither complete nor a circuit with odd number of vertices. It is not bipartite. So* $\chi(\Gamma) \neq 2$. *Obviously* $\chi(\Gamma) \neq 1$. *So,* $\chi(\Gamma) \geq 3$.

Here $\Delta(\Gamma) = 3$, *the maximum degree of all the vertices.*

By Brook's Theorem, $\chi(\Gamma) \leq 3$.

Hence, $\chi(\Gamma) = 3$.

SAMPLE